

Line sets copper; condensate blue dashed (P = pump). · Heads tinted by unit: CU-1 orange, CU-2 blue; shaded rooms are what each head serves. · CU-1 18k · CU-2 36k · cool 21 kBtu/h

draft

M-1 HVAC · 64 Stone Sheep Circle

Ductless: two cold-climate multi-zone mini-split heat pumps with wall heads; the radiant slab (R-1) carries the heat. App version v7. From the owner's plans (Larson House, 6/11/26).

Notes

1. Design conditions: Powell, WY (ASHRAE 2009, Cody Muni AWOS (nearest station to Powell), 99.6 % heating / 1 % cooling). Heating -11.4°F, cooling 87.8°F; inside 70°F / 75°F.
2. No ducts. The radiant slab (R-1) heats every room; the mini-splits cool the rooms with heads and give fast, second-source heat there. Rooms without a head (office, laundry, mud room, powder, halls) take air from the next room through their doors.
3. Sizing: each head carries the cooling load of the rooms it serves plus 15 % for latent and margin, with 6k the smallest head; the living head is held at 9k for the vault, the open kitchen and cooking. Each outdoor unit is the smallest common cold-climate multi-zone with enough ports, with the heads at no more than 130 % of its rating.
4. Heads mount high on the wall, top 6" below the ceiling, on a wall that lets them throw across the room. Exterior-wall heads drain condensate by gravity straight out the wall to a splash block; interior-wall heads (IU-2, IU-3) get a mini condensate pump behind the head that discharges outside with the line set.
5. Line sets: from the unit in a painted line-hide to the wall, through a sealed sleeve about 18" up, inside the wall in their own stud bay to the attic, on the ceiling framing under the attic insulation, then down the wall to the head. The kitchen and living vault has no attic, so nothing crosses it. Pressure test, evacuate and pull a micron vacuum per the maker.
6. Outdoor units on 24" stands above the snow, backs 2'-6" off the wall so they sit outside the drip line of the metal roof (1'-6" eaves); add snow guards on the roof above each unit. Keep 12" behind and 24" in front clear; a wind baffle on the left unit if the yard funnels wind.
7. Each outdoor unit has a dedicated 240 V circuit and a fused disconnect on the wall within sight (DS1, DS2), sized to the nameplate MCA/MOCP once the units are picked (E-1). The heads are powered from their outdoor unit through the 14/4 interconnect run with the line set.

Open items

- Living room head: which outdoor unit feeds it. Drawn on CU-2 (the bedroom unit) because the patio and the vault split the attic: from CU-1 the line set runs about 106 ft around through the front attics, past the 82 ft per-head limit most multi-zone units have.
- Office: no head as drawn (the ask was the right-side bedrooms). A 6k head on the office's right wall would make CU-2 a 5-port.
- Exact models: pick the brand, then confirm the heads are an allowed combination for the outdoor unit and the line lengths are inside its limits.
- Bed walls: the master and Bedroom 1 heads sit on the rear wall between the windows, where a headboard may go. A head over a headboard works with its vanes set level; otherwise it moves to a side wall. Confirm furniture walls before rough-in.
- Ventilation: continuous bath fans (drawn) or paired through-wall ERVs (about 133 CFM continuous under the IRC).
- Mechanical closet: no air handler now, so it is free for the radiant boiler or heat pump and manifold (R-1).

8. Wired wall controllers T1-T6 at 48" by each door; set the heads to cool and to heat only below the slab's setpoint so they don't fight the radiant.
9. Ventilation: with no ducts, the draft assumes continuous low-speed bath fans (exhaust-only, E-1) for the IRC whole-house rate; paired through-wall ERVs are the upgrade (see open items).

Indoor heads

Tag	Room	Unit	Size	Position	Cool Btu/h	Line set	Condensate	Serves
IU-1	Master Bed	CU-1	6k	on the rear wall, centred 7'-1/2" from the left wall; top at 8'-6" (6" below the ceiling)	3,439	1/4" x 3/8" · 59 ft	gravity	Master Bed, Master Bath, Master Closet, Master Hall
IU-2	Dining	CU-1	6k	on the left wall, centred 7'-1/2" from the rear wall; top at 8'-6" (6" below the ceiling)	3,485	1/4" x 3/8" · 55 ft	pump	Dining
IU-3	Living	CU-2	9k	on the right wall, centred 20'-2" from the rear wall; top at 8'-6" (6" below the ceiling)	7,377	1/4" x 3/8" · 38 ft	pump	Living, Kitchen, Entry, Pantry
IU-4	Bedroom 1	CU-2	6k	on the rear wall, centred 6'-6 1/2" from the left wall; top at 8'-6" (6" below the ceiling)	2,263	1/4" x 3/8" · 63 ft	gravity	Bedroom 1, Bedroom 1 Closet, Bath 1
IU-5	Bedroom 2	CU-2	6k	on the right wall, centred 6'-3 1/2" from the rear wall; top at 8'-6" (6" below the ceiling)	2,149	1/4" x 3/8" · 30 ft	gravity	Bedroom 2, Closet (Bedroom 2), Bath 2
IU-6	Bedroom 3	CU-2	6k	on the right wall, centred 4'-3" from the closet wall; top at 8'-6" (6" below the ceiling)	1,797	1/4" x 3/8" · 25 ft	gravity	Bedroom 3, Closet (Bedroom 3)

Outdoor units

Tag	Location	Size	Heads connected	Cool block Btu/h	Heat at 5°F / design	Line sets
CU-1	Left side yard, on the master closet wall	18k, 2 ports	12k (67 %)	6,232	18,000 / 13,500	113 ft, longest 59
CU-2	Right side yard, on the Bedroom 3 closet wall	36k, 4 ports	27k (75 %)	12,227	36,000 / 27,000	156 ft, longest 63

Line sets

Set	Run	Size	Length	Route
LS-1	CU-1 → IU-1	1/4" x 3/8"	59 ft	line-hide to the wall, up inside the wall to the attic, across the attic, down the wall to the head
LS-2	CU-1 → IU-2	1/4" x 3/8"	55 ft	line-hide to the wall, up inside the wall to the attic, across the attic, down the wall to the head
LS-3	CU-2 → IU-3	1/4" x 3/8"	38 ft	line-hide to the wall, up inside the wall to the attic, across the attic, down the wall to the head
LS-4	CU-2 → IU-4	1/4" x 3/8"	63 ft	line-hide to the wall, up inside the wall to the attic, across the attic, down the wall to the head
LS-5	CU-2 → IU-5	1/4" x 3/8"	30 ft	line-hide to the wall, up inside the wall to the attic, across the attic, down the wall to the head
LS-6	CU-2 → IU-6	1/4" x 3/8"	25 ft	line-hide to the wall, up inside the wall to the attic, across the attic, down the wall to the head

Room loads

Room	Floor sf	Heat Btu/h	Cool Btu/h	Head
Master Bed	240	2,785	1,725	IU-1
Master Bath	192	3,455	960	IU-1
Master Closet	96	1,363	271	IU-1
Master Hall	22	108	34	IU-1
Laundry	87	904	587	— (spill from next room)
Powder	30	150	47	— (spill from next room)
Mud Room	133	1,236	356	— (spill from next room)
Dining	136	3,771	3,031	IU-2
Kitchen	370	2,280	2,370	IU-3
Pantry	77	1,441	694	IU-3
Mechanical	25	493	99	— (spill from next room)
Living	524	5,927	3,095	IU-3
Entry	76	1,223	256	IU-3
Coat Closet	6	30	9	— (spill from next room)
Office	134	1,977	1,571	— (spill from next room)
Bedroom Hall	86	422	133	— (spill from next room)
Bedroom 1	176	3,424	1,729	IU-4
Bedroom 1 Closet	18	86	27	IU-4
Bath 1	53	1,443	211	IU-4
Closet (Bedroom 2)	26	521	84	IU-5
Bedroom 2	178	2,363	1,615	IU-5
Bath 2	79	817	170	IU-5
Bedroom 3	169	3,163	1,436	IU-6
Closet (Bedroom 3)	26	916	127	IU-6

System

Item	Size	Basis
Heating load	40,299 Btu/h	70°F inside, -11.4°F outside; carried by the radiant slab (R-1)
Cooling load	20,637 Btu/h sensible	75°F inside, 87.8°F outside
Outdoor units	CU-1 18k + CU-2 36k	cold-climate multi-zone, sized on cooling and the head count
Mini-split heat	40,500 Btu/h at -11°F	backup to the slab in the rooms with heads

Item	Size	Basis
Indoor heads	6	high-wall, 6" below the ceiling

Bill of materials

Rough-order quantities from the layout, for pricing. The HVAC contractor takes off the final list.

Item	Qty	Notes
Outdoor unit, 18k 2-port	1	CU-1, cold-climate multi-zone
Outdoor unit, 36k 4-port	1	CU-2, cold-climate multi-zone
Wall head, 6k	5	
Wall head, 9k	1	
Line set, insulated	297 ft	all sets + 10 %; sizes per the line-set table
Interconnect cable 14/4	297 ft	with the line sets
Line-hide cover and wall sleeves	2 runs + 6 sleeves	painted to the siding
Condensate pump	2	IU-2, IU-3
Wired wall controller	6	T1-T6
Stand, pad, snow guards	2	24" stands
Disconnect	2	DS1, DS2; circuits on E-1